SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 01	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	26/10/2026	**Duration:**	60 minutes

Code of Conduct

I M. Srilakshmi (192525404) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M. Srilakshmi

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**
   - Identify all primary and secondary actors interacting with the system.
   - List the major use cases associated with each actor.
   - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction
   - Problem Description

    - Scope of the System
    - Functional Requirements
    - Non-Functional Requirements
    - Use Case Descriptions
    - Data Requirements
    - External Interface Requirements
    - System Features
5. **Identify Assumptions and Constraints**
    - List the assumptions considered while developing the system.
    - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
    - Verify whether all stakeholder requirements have been addressed.
    - Check for ambiguity, redundancy, inconsistency, and missing requirements.
    - Suggest improvements to enhance the quality and completeness of the requirement specification.

Title: Assess the Accessibility of Roads, Transport Facilities, and Essential Services

1. Study and Analyze the Scenario

1.1 Problem Statement

Accessibility to roads, public transport facilities, and essential services is important for the daily lives of people. However, many people face difficulties in reaching hospitals, schools, bus stops, railway stations, government offices, and other important places due to poor road conditions, a lack of transport facilities, traffic congestion, and insufficient accessibility information.

People with disabilities, elderly people, students, and people living in rural areas may face additional challenges when travelling to these locations.

The proposed system, Road and Essential Services Accessibility Assessment System, helps users identify accessible roads, evaluate transport facilities, locate essential services, and report accessibility problems. It provides information through a digital platform using maps, accessibility ratings, and reports.

1.2 Business Domain

- Domain: Software Engineering, Geographic Information Systems (GIS), Smart City Management.
- Application: Road, transport, and essential services accessibility assessment.
- Technology: Web application, digital maps, database management, GPS/location services, and data analytics.

1.3 Stakeholders

Stakeholder	Role and Responsibility
General Public	Search for accessible routes and essential services.
People with Disabilities	Identify wheelchair-accessible roads, transport, and facilities.
Elderly People	Find convenient routes and nearby services.
Students	Locate schools, colleges, and transport facilities.
Transport Authorities	Provide transport information and improve transport accessibility.
Municipal Authorities	Monitor road conditions and resolve accessibility issues.
Healthcare Providers	Maintain information about hospitals and healthcare facilities.
System Administrator	Manage users, records, and system operations.
Government Officials	Review accessibility reports and plan infrastructure improvements.

1.4 Existing Challenges

1. Lack of accurate information about accessible roads and footpaths.
2. Difficulty identifying wheelchair-accessible public transport.
3. Poor road conditions, potholes, and damaged footpaths.
4. Lack of information about nearby hospitals, schools, and other essential services.
5. Limited information about ramps, lifts, accessible entrances, and parking.
6. Difficulty reporting and tracking accessibility problems.
7. Lack of centralized data for authorities to make improvement decisions.

1.5 Objectives of the Proposed System

- Identify and assess the accessibility of roads and footpaths.
- Provide information about public transport facilities.
- Locate nearby hospitals, schools, pharmacies, and government offices.
- Help users find suitable routes based on their accessibility needs.
- Allow users to report road and transport accessibility problems.
- Generate accessibility ratings, summaries, and reports.
- Help authorities prioritize infrastructure improvements.
- Improve safe, convenient, and inclusive access to essential services.

2. Functional and Non-Functional Requirements

2.1 Functional Requirements

Functional requirements describe what the system must do.

ID	Functional Requirement
FR1	The system shall allow users to register, log in, and manage their profiles.
FR2	The system shall display roads and transport facilities on a digital map.
FR3	The system shall assess road conditions, footpaths, ramps, and pedestrian crossings.
FR4	The system shall display available public transport facilities and their accessibility information.

FR5	The system shall identify nearby hospitals, schools, pharmacies, and government offices.
FR6	The system shall allow users to search for locations and essential services.
FR7	The system shall suggest routes based on accessibility requirements.
FR8	The system shall allow users to submit accessibility complaints with descriptions and photographs.
FR9	The system shall allow authorities to review, update, and resolve reported issues.
FR10	The system shall calculate accessibility scores using defined assessment criteria.
FR11	The system shall generate reports showing inaccessible locations and common problems.
FR12	The system shall allow administrators to manage user accounts, map data, and service information.
FR13	The system shall allow authorized users to view complaint status and assessment results.
FR14	The system shall display the date and source of accessibility information wherever available.

2.2 Non-Functional Requirements

Non-functional requirements describe how well the system must work.

Type	Requirement
Performance	Common searches and map requests should normally respond within 3 seconds under the specified workload.
Security	The system shall protect user information using secure authentication and appropriate access controls.
Reliability	The system shall process and store reports accurately without unintended data loss.
Usability	The interface shall be simple, understandable, and easy to navigate.
Scalability	The system shall support increasing numbers of users, locations, and reports.
Maintainability	The system shall use modular code so features can be updated easily.
Availability	The system should target at least 99% monthly availability, excluding agreed maintenance periods.
Accessibility	The interface should support keyboard navigation, readable text, and screen readers.
Compatibility	The system should work on commonly used desktop and mobile browsers.
Data Accuracy	Accessibility assessments shall use defined criteria and identify outdated or unverified information.
Privacy	The system shall collect only necessary personal information and restrict access to sensitive data.

Note: Performance and availability figures are proposed targets for the project and should be verified during testing.

3. Identify Actors and Use Cases

3.1 Primary and Secondary Actors

Primary actors directly use the system to obtain information or perform tasks.

- General User
- Person with Disability
- System Administrator
- Municipal Authority
- Transport Authority

Secondary actors provide supporting information or services.

- Map/GPS Service
- Database System
- Notification Service
- Essential Service Providers

3.2 Major Use Cases

Actor	Use Cases
General User	Register, log in, search locations, view maps, find essential services, view routes.
Person with Disability	Search accessible routes, check ramps and lifts, view wheelchair-friendly transport facilities.
Municipal Authority	Review road conditions, verify complaints, update issue status, generate reports.
Transport Authority	Update transport accessibility details, review transport complaints.
System Administrator	Manage users, update records, maintain system data, manage access permissions.
Map/GPS Service	Provide map information, location details, and route data.
Essential Service Providers	Provide or update facility details and accessibility information.

3.3 Use Case Diagram

The following diagram represents the interactions between users, authorities, supporting services, and the proposed system.

Road and Essential Services Accessibility Assessment System

Supporting map, database, and notification services supply data to the system. The diagram is a simplified use-case overview.

3.4 Explanation of the Use Case Diagram

1. General User: Searches for locations and essential services and views accessible routes.
2. Person with Disability: Checks wheelchair-friendly roads, ramps, transport facilities, and entrances.
3. Municipal Authority: Reviews road-related complaints and updates their resolution status.
4. Transport Authority: Maintains public transport accessibility information.
5. System Administrator: Manages users, permissions, and system records.

6. Map/GPS Service: Supplies map and location data for route searches.

4. Software Requirement Specification (SRS)

4.1 Introduction

The Software Requirement Specification (SRS) defines the functional and non-functional requirements of the Road and Essential Services Accessibility Assessment System. It describes the features, actors, data, interfaces, and operating conditions required to develop the proposed application.

The system aims to provide reliable accessibility information about roads, public transport facilities, and essential services to help people make informed travel decisions.

4.2 Problem Description

People often lack a single platform that provides detailed accessibility information about roads, footpaths, public transport, hospitals, schools, and other essential services. Existing information may be incomplete, outdated, or difficult to find.

The proposed system addresses this problem by collecting accessibility information, displaying it on digital maps, enabling users to report issues, and helping authorities monitor and improve accessibility.

4.3 Scope of the System

The system includes the following features:

- User registration and authentication.
- Road and footpath accessibility assessment.
- Public transport accessibility information.
- Search for nearby essential services.
- Accessibility-based route suggestions.
- User complaints and issue tracking.
- Accessibility ratings and reports.
- Authority and administrator dashboards.

Out of scope: Direct repair of roads, physical construction of ramps, operation of public transport, and automatic verification of every real-world accessibility condition.

4.4 Functional Requirements

The functional requirements are detailed in Section 2.1. The main system functions are:

1. User management.
2. Road and footpath assessment.
3. Transport facility assessment.
4. Essential service location.
5. Accessibility-based navigation.
6. Complaint management.
7. Accessibility scoring.

8. Report generation.
9. Administrative management.

4.5 Non-Functional Requirements

The system shall provide adequate performance, security, reliability, usability, scalability, maintainability, availability, compatibility, and accessibility as defined in Section 2.2.

These requirements ensure that the application remains useful, secure, understandable, and dependable for different categories of users.

4.6 Use Case Descriptions

Use Case 1: Search for Accessible Routes

Item	Description
Use Case ID	UC-01
Actor	General User / Person with Disability
Objective	Find a route that matches the user's accessibility needs.
Precondition	The system and map service are available.
Main Flow	1. User enters the starting point and destination. 2. User selects accessibility preferences. 3. System retrieves available road and route data. 4. System displays suitable routes and accessibility details.
Alternative Flow	If route data is unavailable, the system informs the user and offers other available information.
Postcondition	The user receives route suggestions with available accessibility details.

Use Case 2: Report an Accessibility Problem

Item	Description
Use Case ID	UC-02
Actor	General User
Objective	Report an inaccessible road, damaged footpath, or transport facility.
Precondition	The user can access the complaint submission feature.
Main Flow	1. User selects a location. 2. User describes the problem. 3. User optionally uploads a photograph. 4. System validates and stores the report. 5. System displays a reference number and current status.
Alternative Flow	If required information is missing, the system asks the user to complete it.
Postcondition	The complaint is stored for review by an authorized authority.

Use Case 3: Manage Accessibility Complaints

Item	Description
Use Case ID	UC-03

Actor	Municipal Authority / Transport Authority
Objective	Review and resolve accessibility complaints.
Precondition	The authority is authenticated and authorized.
Main Flow	1. Authority opens the complaint dashboard. 2. System displays pending reports. 3. Authority reviews and verifies a report. 4. Authority updates its status and adds remarks. 5. System saves the changes.
Alternative Flow	If a report is invalid or duplicated, the authority records the reason and updates its status.
Postcondition	The complaint status and review details are updated.

Use Case 4: Find Essential Services

Item	Description
Use Case ID	UC-04
Actor	General User
Objective	Locate essential services and view their accessibility details.
Precondition	Service location data is available.
Main Flow	1. User selects a service category. 2. User enters a location or uses location services. 3. System searches available records. 4. System displays nearby services and known accessibility features.
Alternative Flow	If no results are found, the system asks the user to change the search area or category.
Postcondition	The user can view relevant service locations and available details.

Use Case 5: Generate Accessibility Reports

Item	Description
Use Case ID	UC-05
Actor	Authorized Municipal / Transport Authority
Objective	Analyze accessibility problems and identify areas for improvement.
Precondition	Assessment data and complaint records exist.
Main Flow	1. Authority selects an area and reporting period. 2. System retrieves relevant records. 3. System calculates summaries and accessibility scores. 4. System displays the report. 5. Authorized user exports the report if supported.
Alternative Flow	If data is insufficient, the system displays a message identifying the missing information.
Postcondition	An accessibility summary is available for planning and review.

4.7 Data Requirements

The system must store and manage the following information.

Data Entity	Important Fields

User	User ID, name, email, password hash, role
Road	Road ID, name, location, surface condition, footpath condition
Transport Facility	Facility ID, type, location, accessibility features
Essential Service	Service ID, name, category, address, contact details, accessibility features
Accessibility Assessment	Assessment ID, location ID, criteria, score, assessment date, verification status
Complaint	Complaint ID, user ID, location, description, photograph reference, status, date
Route	Route ID, starting point, destination, route details, accessibility restrictions
Report	Report ID, reporting period, area, accessibility summary, generated date

Data handling requirements:

- Validate all required fields before saving records.
- Assign unique identifiers to users, locations, assessments, and complaints.
- Restrict access to personal information based on user roles.
- Maintain timestamps for assessment and complaint updates.
- Mark unverified or outdated accessibility information clearly.
- Back up important records according to the system's backup policy.

4.8 External Interface Requirements

A. User Interface

- Simple home page with search and map features.
- Registration and login pages.
- Accessibility filters and service categories.
- Complaint submission form.
- Dashboard for authorities.
- Reports page with summaries and charts.

B. Hardware Interface

- Desktop computers, laptops, tablets, and smartphones.
- GPS-capable devices when location-based functionality is used.
- Internet-connected devices for accessing the application.

C. Software Interface

- Web browser.
- Database management system.
- Digital map and routing service.
- GPS/location API where supported.
- Optional notification service for complaint updates.

D. Communication Interface

- HTTPS for secure web communication.
- Secure API requests between the application and external services.
- Appropriate authentication and authorization for protected operations.

4.9 System Features

Feature	Description	Expected Benefit
Road Accessibility Assessment	Records road quality, footpaths, crossings, and ramps.	Helps users identify accessible paths.
Transport Accessibility	Displays known facilities such as accessible vehicles, stops, lifts, and boarding support.	Supports better transport planning.
Essential Services Locator	Finds hospitals, schools, pharmacies, and public offices.	Reduces difficulty locating services.
Accessible Route Search	Suggests routes based on recorded accessibility conditions.	Helps users choose suitable routes.
Complaint Management	Records, tracks, and updates accessibility issues.	Improves communication with authorities.
Accessibility Scoring	Calculates scores using defined assessment criteria.	Supports comparison of locations.
Reports and Dashboard	Summarizes complaints, scores, and problem areas.	Supports infrastructure planning.
User and Data Management	Controls accounts, permissions, and records.	Improves data protection and administration.

5. Identify Assumptions and Constraints

5.1 Assumptions

The following assumptions are considered while developing the system:

1. Users have access to a smartphone, computer, or other internet-connected device.
2. Users provide correct location details and accurate problem descriptions.
3. Map and GPS services provide usable location information when available.
4. Municipal and transport authorities participate in reviewing accessibility complaints.
5. Information about roads, transport facilities, and essential services can be collected from reliable sources.
6. Users can access the application without specialized technical knowledge.
7. Accessibility scores are calculated using predefined criteria.
8. Accessibility information may change over time and must be updated periodically.
9. Users can report problems even when some location or facility details are unavailable, provided the system can record sufficient information.

5.2 Constraints

Constraints are limitations that may affect the development and operation of the system.

Constraint Type	Description
Technical	The system depends on available map APIs, internet connectivity, and compatible software.
Operational	Authorities must review complaints and update information regularly.
Economic	Development, hosting, map API usage, database storage, and maintenance may require funding.
Legal	The system must follow applicable privacy, data protection, accessibility, and information security requirements.
Time-Related	The project must be completed within the allocated academic project schedule.
Data-Related	Incomplete, outdated, or unverified data may affect accessibility assessments.
Infrastructure	GPS accuracy and internet availability may vary by location.
Resource	The development team may have limited access to real-world road surveys and accessibility experts.

5.3 Technical and Operational Limitations

- The system cannot physically repair damaged roads or construct accessibility facilities.
- Route suggestions depend on the quality and freshness of available map and accessibility data.
- An accessibility score alone cannot guarantee that a route is safe or suitable for every individual.
- Some essential service providers may not supply complete accessibility details.
- User-submitted complaints may require verification before authorities take action.

6. Validate Requirement Completeness and Consistency

Requirement validation ensures that the SRS is complete, understandable, consistent, and suitable for implementation.

6.1 Requirement Validation Checklist

Validation Item	Verification Method	Expected Result
Completeness	Check all stakeholder needs against the requirements list.	All important user and authority needs are covered.
Consistency	Compare functional and non-functional requirements.	No conflicting requirements exist.
Clarity	Review each requirement for clear wording.	Requirements are specific and understandable.
Non-Redundancy	Identify repeated requirements.	Each requirement has a distinct purpose.
Feasibility	Review technology, budget, and project duration.	Requirements can reasonably be implemented.
Testability	Define measurable acceptance criteria.	Each requirement can be verified through testing.

Traceability	Link stakeholder needs to requirements and test cases.	Every major need can be traced to implementation and testing.
Security	Review authentication, permissions, and data handling.	Sensitive information is appropriately protected.
Accessibility	Test keyboard use, screen-reader support, and readable content.	Users with different accessibility needs can use the interface.

6.2 Requirement Completeness

The following checks help determine whether the proposed requirements cover the entire system:

1. Confirm that users can search for roads, transport facilities, and essential services.
2. Confirm that people with disabilities can view relevant accessibility features.
3. Confirm that users can report problems and track complaint status.
4. Confirm that authorized authorities can verify reports and update records.
5. Confirm that accessibility assessments and reports can be generated.
6. Confirm that user authentication, permissions, privacy, and data validation are included.
7. Confirm that errors, missing data, and unavailable external services are handled appropriately.

6.3 Ambiguity, Redundancy, and Inconsistency Checks

Potential Issue	Example	Suggested Improvement
Ambiguity	“The system should be fast.”	Define a measurable response-time target and test conditions.
Ambiguity	“Show accessible roads.”	Specify the accessibility criteria and information displayed.
Redundancy	Two requirements describe the same complaint submission function.	Combine them into one clearly defined requirement.
Inconsistency	Users are allowed to edit verified assessments while the system says only authorities can edit them.	Define separate permissions for users and authorities.
Missing Requirement	No requirement describes outdated accessibility information.	Add assessment dates and verification status.
Missing Requirement	No handling is defined for map service failure.	Display an appropriate error message and preserve other available functions.
Missing Requirement	No protection is specified for uploaded photographs.	Validate uploads and restrict access to protected files.

6.4 Suggested Improvements

- Assign a unique ID to every functional and non-functional requirement.
- Define a standard accessibility assessment checklist for roads and facilities.
- Record the date, source, and verification status of accessibility information.
- Introduce role-based permissions for users, administrators, and authorities.
- Add duplicate complaint detection and complaint status notifications.
- Define measurable performance, availability, and security targets.

- Conduct usability testing with people with disabilities and other representative users.
- Maintain a requirement traceability matrix linking requirements, use cases, and test cases.

6.5 Requirement Traceability Matrix

Stakeholder Need	Requirement ID	Related Use Case	Testing Method
Find accessible routes	FR7	UC-01	Route search testing
Report road problems	FR8	UC-02	Complaint submission testing
Resolve accessibility issues	FR9	UC-03	Authority workflow testing
Find essential services	FR5, FR6	UC-04	Search and location testing
Analyze accessibility problems	FR10, FR11	UC-05	Report and scoring validation
Protect user information	NFR Security, Privacy	User authentication and administration	Security testing

6.6 Final Validation Outcome

The proposed SRS covers the main requirements of the Road and Essential Services Accessibility Assessment System. It defines the problem, objectives, stakeholders, functional and non-functional requirements, actors, use cases, data requirements, interfaces, assumptions, and constraints.

Before implementation, the requirements should be reviewed by representative users and authorities. Any missing requirements should be documented, and measurable acceptance criteria should be established for testing.

Conclusion

The Road and Essential Services Accessibility Assessment System aims to improve access to roads, public transport, hospitals, schools, and other essential services through a centralized digital platform. It provides accessibility assessments, route information, complaint management, and reports for infrastructure planning. By following the proposed SRS, the development team can build a more usable, secure, reliable, and inclusive system that supports informed travel decisions and helps authorities identify accessibility improvements.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25